

Cell Cycle and Cell Division

1. Match the following and choose the correct option.

Column-I	Column-II
A. Karyokinesis	i. Furrow
B. Cytokinesis	ii. Cell plate formation
C. Plant cell	iii. Cytoplasm division
D. Animal cell	iv. Nuclear division

Choose the most appropriate answer

- (1) A-iv, B-iii, C-ii, D-i
- (2) A-iii, B-iv, C-i, D-ii
- (3) A-ii, B-i, C-iv, D-iii
- (4) A-iv, B-ii, C-iii, D-i

2. Given below are two statements;

Statement-I : Centriole begin to move towards opposite poles of the cell in anaphase.

Statement-II: Condensation of chromosomes is completed in metaphase.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II are correct.
- (4) Both Statement-I and statement-II are correct.

3. Yeast for example can progress through the cell cycle in only about.....

- (1) 40 minutes
- (2) 90 minutes
- (3) 1440 minutes
- (4) 60 minutes

4. Given below are two statements :

Assertion (A) : Plant cytokinesis takes place by a different mechanism than that of animal cytokinesis.

Reason (R) : Plant cells are enclosed by a relatively inextensible cell wall.

Choose the most appropriate answer

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

5. Identify the incorrect statement about G₁ -phase

- (1) It occurs between M and S phase
- (2) In it, cells are metabolically active and continuously grow
- (3) It is pre-synthetic or post mitotic phase
- (4) In this phase DNA replicates

6. Which of the following is not related to mitosis?

- (1) It is an equational division.
- (2) It involves recombination and crossing over.
- (3) Sister chromatids are separated during Anaphase.
- (4) Nuclear membrane disappears during late prophase.

7. Interkinesis is gap between –

- (1) Between prophase I and Metaphase I
- (2) Two mitotic divisions
- (3) Anaphase I and Anaphase II
- (4) Telophase I and Prophase II

8. Which of the following option correctly describes the cell as shown in figure ?



- (1) 8 chromosomes ; 8 homologous pair
(2) 16 chromosomes ; 4 homologous pair
(3) 8 chromosomes ; 4 homologous pair
(4) 16 chromosomes ; 8 homologous pair
9. Arrange the following events of meiosis in the correct sequence –

- A. Terminalisation of chiasmata
B. Crossing over
C. Synapsis
D. Disjunction of chromosomes
E. Dissolution of synaptonemal complex

The correct sequence is-

- (1) A → B → C → D → E
(2) E → D → C → B → A
(3) C → B → D → E → A
(4) C → B → E → A → D

10. Crossing over takes place in :-

- (1) Zygotene
(2) Pachytene
(3) Diplotene
(4) Diakinesis

11. What is the significance of meiosis :-

- (1) The conservation of specific chromosome number of each species is achieved across generations in sexually reproducing organisms.
(2) It increases the genetic variability in the population of organisms from one generation to the next.
(3) Meiosis decreases the genetic variability in the population of organisms from one generation to next.
(4) 1 and 2 option is correct.

12. Given below are two statements:

Assertion (A) : Meiosis results in the reduction of chromosome number by half in sexually reproducing organisms.

Reason (R) : This reduction is essential to maintain the specific chromosome number when haploid gametes combine during fertilization to form a diploid zygote, ensuring genetic stability across generations.

Choose the most appropriate answer from the options given below:

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
(3) Assertion (A) is true, Reason (R) is false.
(4) Assertion (A) is false, and Reason (R) is true

13. Mark the given statements as true or false and choose the correct option.

- (a) Cell growth results in disturbing the ratio between the nucleus and the cytoplasm.
(b) At the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between the two daughter cells.
(c) Cytokinesis takes place before karyokinesis.
(d) The process of cytokinesis causes division of Parental cell nucleus into two daughter nuclei.

- (1) (a) True, (b) True, (c) False, (d) False
(2) (a) True, (b) True, (c) True, (d) False
(3) (a) True, (b) True, (c) True, (d) True
(4) (a) False, (b) False, (c) True, (d) True

14. Which of the following is smallest phase of the cell cycle ?

- (1) Prophase
(2) Interphase
(3) Anaphase
(4) Metaphase

15. A cell completes its cell cycle in 50 minutes. The approximate duration of the M-phase will be

- (1) 2.5 Minutes
- (2) 45 Minutes
- (3) 5 Minutes
- (4) 25 Minutes

16. Given below are the two statements:

Statement-I : Nuclear membrane & Nucleolus reappear in telophase.

Statement-II : During metaphase, the chromosomes are arranged on the equatorial plate.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are correct.
- (2) Statement I is correct but statement II is incorrect.
- (3) Statement I is incorrect but statement II is correct.
- (4) Both statement I and II are incorrect.

17. Refer to the following statements describing Prophase I of meiosis and answer the question.

- (i) Thin thread like chromosomes
- (ii) Appearance of recombination nodules
- (iii) Formation of bivalents/tetrads
- (iv) Terminalisation of chiasmata
- (v) Appearance of chiasmata

Arrange the given statements in the correct sequence their occurrence during prophase I

- (1) (i) → (iii) → (ii) → (iv) → (v)
- (2) (i) → (ii) → (iii) → (iv) → (v)
- (3) (i) → (iv) → (v) → (ii) → (iii)
- (4) (i) → (iii) → (ii) → (v) → (iv)

18. The major difference between meiosis-I and meiosis-II is that, the

- (1) The number of chromosomes remains the same in former while it reduces to half in latter.

- (2) The number of chromosomes reduces to half in the former while it remains the same in latter.
- (3) The number of chromosomes remains the same in both cases.
- (4) The number of chromosomes reduces to half in both cases.

19. Identify the correct statement.

- (1) Meiosis involves two sequential cycles of nuclear and cell division but only a single cycle of DNA replication
- (2) Meiosis-I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S-phase.
- (3) Four haploid cells are formed at the end of meiosis-II
- (4) All of the above.

20. Given below are two statements:

Statement-1 : In meiosis, DNA replication occurs once whereas nuclear division occurs twice.

Statement-2 : Crossing over occurs between non-sister chromatids of non homologous chromosomes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-1 is true and statement-2 is correct explanation for statement-1
- (2) Statement-1 is true and statement-2 is also true but statement-2 is not correct explanation of statement-1
- (3) Statement-1 is true but statement-2 is false
- (4) Statement-1 and statement-2 both are false.

21. Prophase I is further divided into five phases based on

- (1) Chromosomal behaviour
- (2) Chromosomal morphology
- (3) Chromosomal number
- (4) Chromosomal shape.

22. Meiosis-II is initiated immediately after _____

usually before the _____ have fully elongated.

- (1) Karyokinesis, chromosomes
- (2) Cytokinesis, Genes
- (3) Cytokinesis, nucleus
- (4) Cytokinesis, chromosomes

23. Given below are the two statements:

Assertion : Meiosis is essential for maintaining specific number of chromosome in a species across generations.

Reason : Meiosis increase genetic variability in sexual reproducing organisms.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

24. Mark incorrect statements :-

(A) Meiosis involves only a single cycle of DNA replication

(B) Four haploid cells are formed at the end of meiosis-I

(C) Meiosis occurs in diploid cells

(D) In yeast, cell cycle takes about 90 minutes.

- (1) A and B
- (2) A and C
- (3) Only B
- (4) All are correct

25. Match the following with respect to meiosis:

- | | |
|-----------------------|----------------------------|
| (a) Zygotene | (i) Terminalization |
| (b) Pachytene | (ii) Chiasmata |
| (c) Diplotene | (iii) Crossing over |
| (d) Diakinesis | (iv) Synapsis |

Select the correct option from the following:

(a) (b) (c) (d)

- (1) (ii) (iv) (iii) (i)
- (2) (iii) (iv) (i) (ii)
- (3) (iv) (iii) (ii) (i)
- (4) (i) (ii) (iv) (iii)

26. Read the following features and identify the phase in which it occurs,

(1) Phase of differentiation.

(2) Cells of the phase are metabolically active but no longer proliferate.

(3) Shown by heart cells.

Choose the most appropriate answer from the options given below:

- (1) G₀ phase
- (2) G₁ phase
- (3) S - phase
- (4) G₂ phase

27. Electron micrographic view of which stage of prophase-I will show clear tetrad.

- (1) Leptotene
- (2) Pachytene
- (3) Zygotene
- (4) Diplotene

28. The characteristic feature of diplotene stage of meiosis-I is recognized by.

- (1) Dissolution of synaptonemal complex
- (2) Homologous chromosomes of bivalent start separate from each other
- (3) X-shaped structure chiasmata formed
- (4) All of the above

29. In some organisms karyokinesis is not followed by cytokinesis as a result of which.

- (1) Uninucleate condition arises leading to the formation of syncytium
- (2) Multinucleate condition arises leading to the formation of syncytium
- (3) Liquid endosperm in coconut
- (4) Both 2 and 3

30. Which of the following is correct about G1 phase.

- i. Cell is metabolically active
- ii. Continuous growth of the cell
- iii. Replicate its DNA

- (1) i and ii are correct
- (2) only ii correct
- (3) i, ii and iii are correct
- (4) i and iii are correct

31. During which phase(s) of cell cycle, amount of DNA in a cell remains at 4C level if the initial amount is denoted as 2C?

- (1) G0 and G1
- (2) G1 and S
- (3) Only G2
- (4) G2 and M

32. When does prophase II start usually?

- (1) Chromosomes are fully elongated
- (2) Before chromosomes are fully condensed
- (3) Before chromosomes are fully elongated
- (4) After chromosomes are fully elongated

33. Select the incorrect statements.

- (1) In some social insects haploid cell divide by mitosis to form gametes.
- (2) Mitotic division take place only in the apical meristem
- (3) Mitosis helps to restore the nucleo cytoplasmic ratio
- (4) Mitosis helps in cell repair.

34. How many are associated with meiosis.

- a. Occurs in haploid cell
- b. Occurs in diploid cell
- c. Pairing of homologous chromosomes
- d. Reductional division
- e. Spindle fibres are absent

- (1) 2
- (2) 3
- (3) 4
- (4) 1

35. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R)

Assertion : Crossing over take place during pachytene stage.

Reason : It is process of interchange of chromatid between non sister chromatid of homologous chromosomes.

In the light of the above statements, Choose the most appropriate answer from the options given below :

- (1) Both Assertion & Reason are True and the Reason is the correct explanation of the Assertion
- (2) Both Assertion & Reason are True but the Reason is not the correct explanation of the Assertion
- (3) Assertion is True statement but Reason is false
- (4) Both Assertion and Reason are False statements

36. Cells at the end of prophase, when viewed under the microscope do not show_____.

- (1) Golgi complexes, endoplasmic reticulum, nucleolus and the nuclear envelope
- (2) Only nuclear envelope
- (3) Only endoplasmic reticulum
- (4) Only nucleolus

37. Arrange the following stages of mitotic cell cycle in correct order and choose the correct option.

- (1) M phase – G1 phase - S phase – G2 phase
- (2) S phase – G2 phase – G1 phase - M phase
- (3) S-phase - M phase - G2 phase- G1 phase
- (4) S phase – G1 phase - M phase – G2 phase

38. Identify the incorrect statement about Quiescent stage of cell cycle

- (1) The cells that do not divide further exit G1 phase
- (2) Cells in this stage become metabolically inactive
- (3) Cells in this stage no longer proliferate unless called on to do so
- (4) Quiescent stage is also known as G0 phase

39. Given below are two statements:

Statement-I : If the cell has diploid or ' $2n$ ' number of chromosomes at G1 then after 'S' phase the number of chromosomes will be double ($4n$).

Statement-II : During prophase, protein are synthesized in preparation for mitosis.

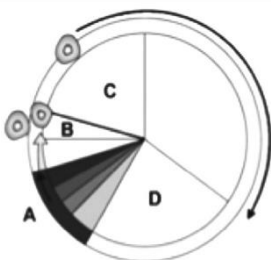
In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement II are correct.
- (4) Both Statement-I and statement-II are correct.

40. Find the incorrect statement.

- (1) M-Phase involves a major reorganization of virtually all components of the cell.
- (2) Cell-division is a progressive process and very clear-cut lines cannot be drawn between various stages.
- (3) Prophase is marked by the initiation of condensation of chromosomal material
- (4) Microtubules, the carbohydrate component of the cell cytoplasm help in mitosis.

41. Identify the mismatched pair.



- (1) A-Starts with karyokinesis and ends with cytokinesis
- (2) B-Stage where cells are inactive metabolically
- (3) C-Cell grows and carries out normal metabolism
- (4) D-Period of cytoplasmic growth

42. Given below are two statements:

Assertion : Amount of DNA get doubled in 'S' phase of interphase.

Reason : DNA replication occur in 'S' phase of interphase.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

43. Which of the following is the most dramatic period of the cell cycle?

- (1) G1- phase
- (2) G2- phase
- (3) S- phase
- (4) M- phase.

44. Given below are two statements:

Assertion (A) : Morphology of chromosomes is most studied during metaphase of cell division.

Reason (R) : Complete disintegration of nuclear envelope marks the end of metaphase, second phase of mitosis.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

45. In some organisms, such as certain fungi and algae, cells undergo mitosis repeatedly without subsequently undergoing cytokinesis. What would result from this?

- (1) A decrease in chromosome number
- (2) Inability to duplicate DNA
- (3) Division of the organism into many cells, most lacking nuclei
- (4) Large cells containing many nuclei

46. Arrange the following events of meiosis in correct order.

- (i) Formation of bivalents
- (ii) Compaction of chromosomes
- (iii) Separation of sister chromatids
- (iv) Crossing over
- (v) Movement of chiasmata towards ends of homologous chromosomes.
- (vi) Formation of Dyad chromosomes

Identify the correct sequence.

- (1) ii, i, iv, v, vi, iii
- (2) ii, iv, i, v, vi, iii
- (3) i, ii, v, iv, iii, vi
- (4) ii, i, v, iv, vi, iii

47. Given below are two statements;

Statement-I : Chromosomes decondense in telophase I to form chromatin to extended state of interphase nucleus.

Statement-II : Chromosomes recondense in prophase II.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement-I and Statement-II are correct.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Both Statement-I and Statement-II are incorrect.

48. After meiosis I, chromosome number

- (1) is reduced to half
- (2) doubles up
- (3) remains the same
- (4) either (1) or (2)

49. Given below are two statements :

Assertion (A) : In coconut, liquid endosperm represents syncytium.

Reason (R) : In coconut liquid karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises in endosperm.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct but (R) is the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) and (R) are correct and (R) is not the correct explanation of (A).

50. Identify the correct sequence of events during

Prophase I of meiosis :

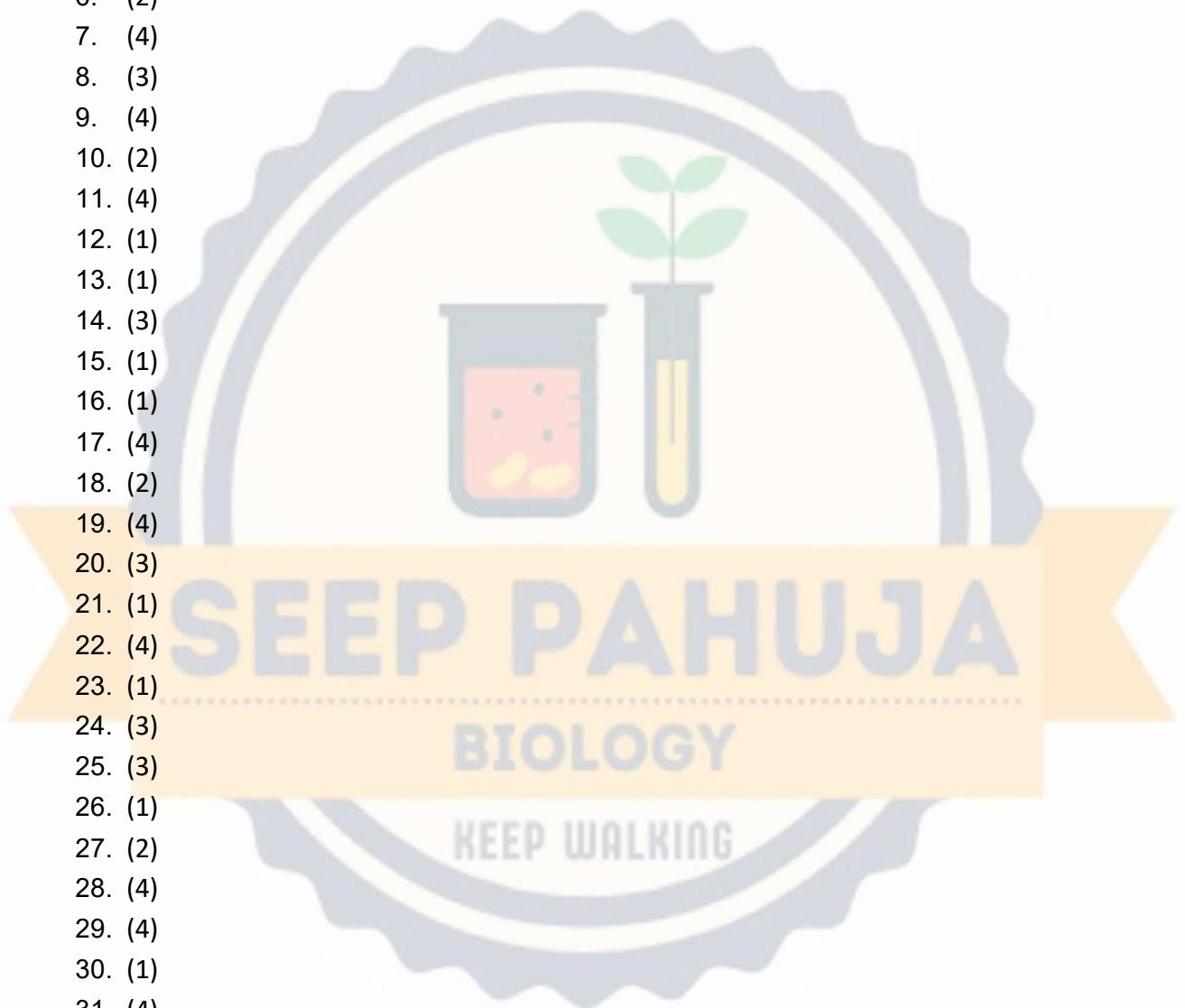
- (a) Synapsis of homologous chromosomes
- (b) Chromosomes become gradually visible under microscope
- (c) Crossing over between non-sister chromatids of the homologous chromosomes
- (d) Terminalisation of chiasmata
- (e) Dissolution of synaptonemal complex

Choose the correct answer from the options given below :

- (1) (a), (b), (c), (d), (e)
- (2) (b), (c), (d), (e), (a)
- (3) (b), (a), (c), (e), (d)
- (4) (a), (c), (d), (e), (b)

Answer Key

1. (1)
2. (3)
3. (2)
4. (1)
5. (4)
6. (2)
7. (4)
8. (3)
9. (4)
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11. (4)
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40. (4)



- 41. (1)
- 42. (4)
- 43. (4)
- 44. (3)
- 45. (4)
- 46. (1)
- 47. (2)
- 48. (1)
- 49. (1)
- 50. (3)

